

# Contributors Guide & DSP Knowledge Base

Architecture specifications, development standards, and canonical signal processing literature.

## System Architecture & Decoupled Modular Design

Overtune 3 is architected into two decoupled subsystems to maximize mathematical reproducibility and audio performance:

1. **dsp/ Core Library:** Pure, standalone C++20 digital signal processing engine. Zero dependencies on GUI frameworks or heavy external libraries. Standard C++ standard library only (`<complex>`, `<vector>`, `<cmath>`).
2. **app/ Application Layer:** Hardware-accelerated Qt 6.4+ / QML desktop application featuring interactive pole-zero drag manipulators, real-time audio playback, and responsive dark/light theme switching.

## Repository Layout

```
overtune3/  
|-- dsp/  
| |-- include/dsp/ # Public DSP headers (Biquad, FilterTypes)  
| |-- src/ # C++20 implementations (Butterworth, Chebyshev, etc.)  
|-- app/  
| |-- src/ # Qt controllers (FilterEngine, Models, Theme)  
| |-- qml/ # Declarative QML controls, pages, and components  
| |-- resources/ # Vector assets and bundled fonts  
|-- latex/ # LaTeX documentation sources and compiled PDFs
```

## Core DSP Contribution Principles

When extending or adding filter topologies to `dsp/`:

- **Deterministic Inner Loops:** Inner audio sample execution loops must operate strictly on pre-allocated buffers with zero heap allocation.
- **Second-Order Section (SOS) Decomposition:** Decompose high-order polynomials into cascaded bi-quads to avoid coefficient sensitivity and numerical instability.
- **Exact Tangent Frequency Pre-Warping:** Always pre-warp analog cutoffs via  $\Omega = 2F_s \tan(\pi F_c / F_s)$  prior to bilinear transform substitution.
- **SIMD Auto-Vectorization:** Structure memory layout to allow compiler auto-vectorization over contiguous sample blocks.

## Local Build & Testing Workflow

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To clone, compile, and run Overtune 3 locally:

```
git clone https://github.com/shadcy/overtune3.git
cd overtune3
cmake -B build -DCMAKE_BUILD_TYPE=Release
cmake --build build --parallel
./build/bin/FilterDesigner
```

## Canonical DSP Reference Literature

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The mathematical foundations implemented across Overtune 3 derive from standard DSP literature:

1. **Alan V. Oppenheim & Ronald W. Schaffer:** Discrete-Time Signal Processing, 3rd Edition, Prentice Hall. (Foundational derivations of the Bilinear Transform, Z-transforms, and IIR stability criteria).
2. **John G. Proakis & Dimitris G. Manolakis:** Digital Signal Processing: Principles, Algorithms, and Applications, Pearson. (Analog prototype synthesis for Butterworth, Chebyshev, and Elliptic filters).
3. **Udo Zölzer:** DAFX: Digital Audio Effects, 2nd Edition, John Wiley & Sons. (Practical audio filter implementations, state-variable models, and Direct Form structures).
4. **Robert Bristow-Johnson:** Cookbook formulae for audio equalizer biquad filter coefficients. (Standard reference for second-order parametric audio shelving and peaking filters).

## Community & External Links

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- [Overtune 3 Project GitHub Repository](#)
- [Issue Tracker & Feature Proposals](#)
- [Maintainer GitHub Profile \(shadcy\)](#)
- [SciPy Signal Processing Documentation](#)
- [Digital Signal Processing Overview – Wikipedia](#)
- [Digital Biquad Filter Topologies – Wikipedia](#)